

Sorting

TABLE OF CONTENTS

1. Understand sorting
2. Few problems on sorting
3. 2 sorting algorithms
 - 3.1 Selection Sort
 - 3.2 Insertion Sort



Notes



Sorting → Arrange data acc to some parameters

Ex → 1, 10, 15, 25, 35 (sorted acc to value)

num-
factors 1, 13, 9, 6, 12 } sorted acc to
 ↓ ↓ ↓ ↓ ↓ num of factors
 1 2 3 4 6

Why sorting?

Helps in organizing, analysing & searching
data Potato

How to sort

Java Arrays.sort(arr) / Collections.sort(arr)
 array arraylist

C++ sort(a, a+n)

Python arr.sort()

**Question (Elements Removal)**

Given N elements, at every step remove an array element.

Cost to remove an element = Sum of array of elements present in an array

Find minimum cost to remove all elements.

NOTE : First add the cost of removal and then remove it.

arr - [2, 1, 4]		2, 1, 4	7
<u>2, 4</u>	7	2, 1	3
4	6	1	1
[]	4	[]	<u>11</u>
	<u>17</u>		

each step, remove biggest number

4, 6, 1	11	3, 5, 1, -3	→ 6
4, 1	5	3, 1, -3	→ 1
1	1	1, -3	-2
[]	<u>17</u>	-3	<u>-3</u>
		[]	<u>2</u>



$a[0]$ $a[1]$ $a[2]$ $a[3]$ } $a < b < c < d$
 $a[0] + a[1] + a[2] + a[3]$
 $a[0] + a[1] + a[2]$
 $a[0] + a[1]$
 $a[0]$

 $4a[0] + 3a[1] + 2a[2] + a[3]$

$$arr[i] * (n-i)$$

Arrays.sort(arr) } $n \log n$

ans = 0

for (i=0; i<n; i++) {

ans += (n-i) * arr[i]

}

return ans

TC: $O(n \log n)$ SC: $O(1)$

**Question (Noble Integers)** { Distinct data }

Given N array elements, calculate number of noble integers.

An element ele in $arr[]$ is said to be noble if { count of smaller elements = ele itself }

$arr = [1, -5, 3, 5, -10, 4]$ $ans = 3$
 $\times \quad \times \quad \checkmark \quad \checkmark \quad \times \quad \checkmark$

$arr = [-3, 0, 2, 5]$ $ans = 1$
 $\times \quad \times \quad \checkmark \quad \times$

Brute : N^2 loop idea

(for each i , count $a[j] < a[i]$
 if count == $a[i]$
 $ans++$

	0	1	2	3
	-3	0	2	5
		<u>0</u>	<u>2</u>	<u>5</u>
smaller	0	1	2	3



Arrays.sort(arr) // sort by value in ascending order

ans = 0

for (i = 0; i < n; i++) {

if (arr[i] == i)

ans++

}

return ans

TC: $O(n \log n)$

SC: $O(1)$



Question (Noble Integers) : { Data can repeat }

arr - [-10 , 1 , 1 , 3 , 100]

small
0 1 2 3 4
0 1 1 3 4

ans = 3

arr - [-10 , 1 , 1 , 2 , 4 , 4 , 4 , 8 , 10]

small
0 1 2 3 4 5 6 7 8
0 1 1 3 4 4 4 7 8
✓ ✓ ✓ ✓ ✓

ans = 5

arr - [-3 , 0 , 2 , 2 , 5 , 5 , 5 , 5 , 8 , 8 , 10 , 10 , 10 , 14]

0 1 2 3 4 5 6 7 8 9 10 11 12 13
0 1 2 2 4 4 4 4 8 8 10 10 10 13
✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓

ans = 7

</> Code

Arrays.sort(arr)

count = 0 , ans = 0

if (arr[0] == 0)
ans++

for (i = 1 ; i < n ; i++) {
if (arr[i] != arr[i-1])
count = i
if (count == arr[i])
ans++
}

TC : $n \log n$
SC : $O(1)$



Selection Sort

idea : Select the minimum element and send that elements to correct position by swapping.

3 5 1 4 2

=> 1 5 3 4 2

=> 1 2 3 4 5

Select smallest guy & get at
current position

2 ~~5~~ 4 ~~6~~ 4 ~~6~~ 5 ~~2~~ ~~5~~ 6





</> Code

```
for (i = 0 ; i < n ; i++) {
```

```
    min_index = i
```

```
    for (j = i + 1 ; j < n ; j++) {
```

```
        if (a[j] < a[min_index]) {
```

```
            min_index = j
```

```
        }
```

```
    }
```

```
    swap(a[i], a[min_index])
```

```
}
```

40

TC: $O(n^2)$

0

1

2

3

20

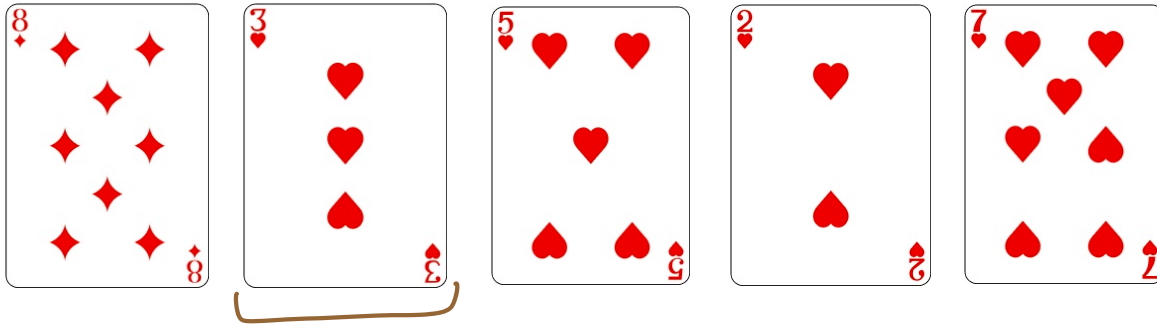
~~50~~~~60~~~~40~~~~20~~~~50~~~~60~~
50

60





Insertion Sort (Arrangement of playing cards)



2 3 5 7 8

Code wise

8 3 5 2 7

3 8 5 2 7

3 5 8 2 7

3 5 2 8 7

3 2 5 8 7

2 3 5 8 7

2 3 5 7 8



</> Code

```
for ( i = 1; i < n; i++) {
```

```
    cur = arr[i]
```

```
    j = i - 1
```

```
    while (j > 0 && arr[j] > cur) {
```

```
        arr[j+1] = arr[j]
```

```
        j--
```

```
    }
```

```
    arr[j+1] = cur
```

```
}
```

0	1	2	3	4
8	3	5	2	7
3	2	2	2	8
	8	5	7	↑
2	3			

0	1	2	3	4
8	3	5	2	7